

Eshan-Powerflow

Automated Document Routing & Approval System

Power Automate | SharePoint | Teams | Excel | AI Builder

Built by: Eshan | Domain: Import-Export & Logistics | Year: 2026

1. Project Overview

EshanPowerflow is a fully automated document routing and multi-stage approval system built in Microsoft Power Automate, purpose-designed for logistics and import-export operations. It eliminates manual handoffs across SharePoint, Microsoft Teams, and Excel — replacing a process that previously required 3–5 hours of daily manual coordination.

PROBLEM	Import-export teams receive high-volume shipment documents (invoices, BLs, customs filings) that must be routed to different departments, reviewed, approved, and logged — manually. Errors cause compliance risk and shipment delays.
SOLUTION	A scheduled Power Automate flow that detects new uploads, routes files based on document type, triggers multi-approver workflows via Teams, logs outcomes in Excel, and moves files to the correct folder — zero manual steps required.

2. Tools & Technology Stack

Tool	Component	Purpose in This Flow
Power Automate	Cloud Flow + AI Builder	Core automation engine; condition logic, approvals, loops
SharePoint	Document Libraries (Uploads, Inward, OUT, Temp)	File storage and folder-based routing trigger

Microsoft Teams	Chat / Channel Messages	Approval notifications and status updates to stakeholders
Excel (Online)	Log Table	Structured audit trail — timestamp, file name, decision, approver
SharePoint List	Free / Process Items	State management — tracks which documents are in-progress

3. Automation Flow Logic

The flow operates on a scheduled recurrence trigger and executes a multi-stage pipeline across three logical phases:

Phase 1 — Trigger & File Detection

The flow is initiated by a recurrence schedule. It scans the SharePoint 'Uploads' folder for new files and evaluates a Condition (PDF Available) to determine whether processing should begin or the run should terminate early

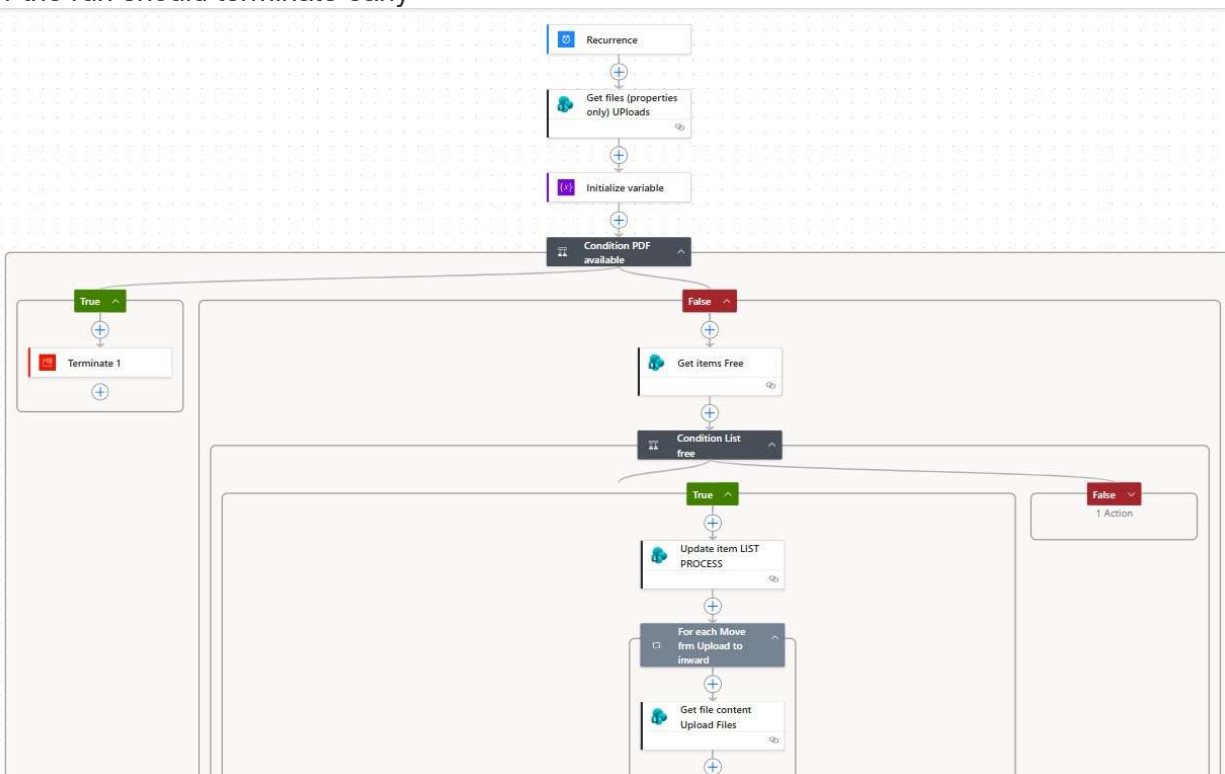


Figure 1: Flow trigger (Recurrence), file detection (Get Files – Uploads), initialization, and Condition PDF Available gate

Phase 2 — File Processing & Movement

When a valid file is detected, the flow reads the SharePoint List to check processing availability (Condition List Free). If the slot is free, it updates the list status, retrieves file content from the Uploads folder, creates a copy in the Inward staging area, and deletes the original. It then fetches files from the IND folder, applies a Select and Join transformation to compile a file manifest, and escalates to approval.

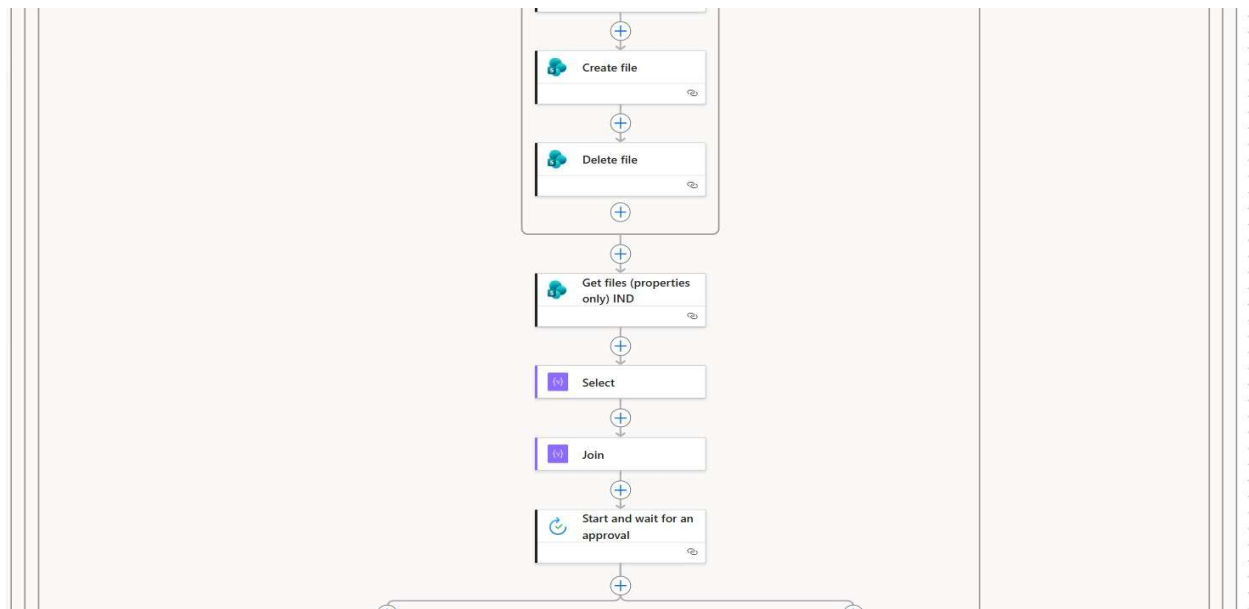


Figure 2: File copy (Create File), cleanup (Delete File), IND folder scan, Select/Join transformation, and Approval trigger

Phase 3 — Approval Routing & Outcome Logging

The approval step pauses the flow and sends a notification via Microsoft Teams to the designated approver(s). Based on the response, a Condition branches into two outcomes. The True path executes file movement to the OUT folder, notifies the team via Teams message, and updates the item status. The False path loops through files using a nested For Each and logs each record as a new row in the Excel audit table.

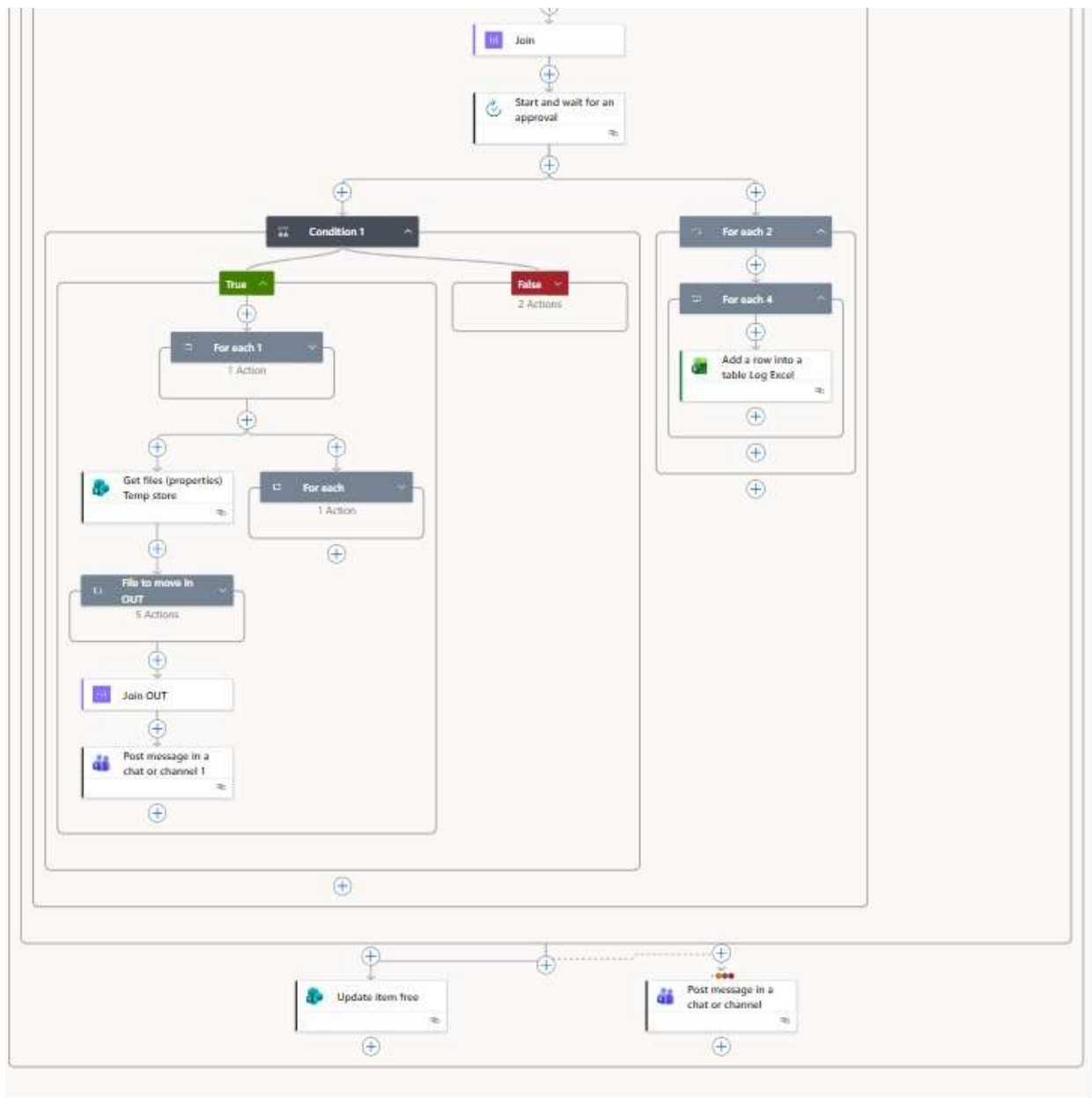


Figure 3: Post-approval branching — True path (file to OUT, Teams notification) and False path (Excel log row per file)

4. Step-by-Step Flow Walkthrough

#	Step / Action	Description
1	Recurrence Trigger	Scheduled polling — flow activates at defined intervals (e.g., every 15 min or hourly)
2	Get Files – Uploads	Retrieves all files currently in the SharePoint Uploads library
3	Initialize Variable	Sets a tracking variable for process state management
4	Condition: PDF Available	Checks if any file meets the criteria (PDF type); terminates run if none found
5	Get Items (List Free)	Reads the SharePoint processing queue list to check slot availability
6	Condition: List Free	Ensures no concurrent run is in progress — prevents race conditions
7	Update List: PROCESS	Marks the list item as 'in-process' to lock the queue
8	For Each – Move Upload to Inward	Iterates over files; gets content, creates copy in Inward, deletes from Uploads
9	Get Files – IND	Retrieves file list from the IND processing folder for manifest compilation
10	Select + Join	Transforms file array into a formatted, comma-separated manifest string
11	Start and Wait for Approval	Sends approval request to designated approver(s) via Teams; flow pauses here
12	Condition 1 (Approval Result)	Branches on Approve/Reject outcome from the approver
13a	True Path – File to OUT	Moves file to OUT folder, posts Teams notification to channel, updates item status
13b	False Path – Log to Excel	Loops through rejected files; adds each as a new row in the Excel audit log table
14	Update Item Free	Releases the process lock — queue slot is freed for next run

15	Post Final Status to Teams	Sends a summary message to the channel regardless of approval outcome
-----------	-----------------------------------	---

5. Business Impact & Results

Metric	Outcome	Business Impact
Manual coordination time eliminated	3–5 hrs/day	Operations team freed for value-added tasks vs. file-chasing
Document misrouting risk	Near zero	Condition logic enforces routing rules; no human judgment required
Approval visibility	Real-time (Teams)	Stakeholders see approval status instantly, not at end-of-day email
Audit trail completeness	100% logged	Every file decision recorded in Excel with timestamp and approver
Concurrent run safety	Race-condition proof	SharePoint list-based locking prevents duplicate processing
Scalability	Template-based	New document types can be added by modifying Condition logic only

6. Portfolio Access Note

NOTE	Power Automate flows run inside an organizational Microsoft 365 tenant and cannot be shared via a public URL. This document, along with annotated screenshots and a GIF walkthrough in the GitHub repository, serves as the portfolio artifact.
-------------	---

How to View This Work

- GitHub Repository: Screenshots (PNG) + this case study document (PDF/DOCX)
- GIF Walkthrough: Recorded screen capture of the live flow execution — see /assets folder in the repo
- Flow Export: The .zip export file (EshanPowerflow_20260604181708.zip) is included in the repo for import into any Power Automate-enabled tenant
- Live Demo: Available on request — can be demonstrated in a screen-share session during interviews

IMPORT INSTRUCTIONS

To test the flow: Download the .zip file → Power Automate → My Flows → Import → Upload .zip → Map connections (SharePoint, Teams, Excel) → Enable the flow. Requires Microsoft 365 license with Power Automate standard connectors.

7. Skills Demonstrated

Technical Skills	Supply Chain / Business Skills
• Power Automate (Cloud Flows, Conditions, Loops) • SharePoint Lists & Document Libraries • Multi-stage Approval Workflows • Excel Online (Table row operations) • Data transformation (Select, Join actions) • Teams integration for real-time notifications	• Document workflow design for import-export ops • Compliance-focused audit trail design • Process bottleneck identification and elimination • Concurrent process safety (queue locking) • Stakeholder communication automation • Supply chain transformation via low-code tooling

EshanPowerflow | Built as part of AI in Supply Chain portfolio transition | Eshan, MBA (Operations, SCM, International Business)